

Shrimaan Yashraj

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ABOUT ME

Driven Mechanical Engineering student with a proven record of designing, simulating, and prototyping innovative solutions for complex mechanical systems, evidenced by multiple national competition wins and a patent application. As Technical Head of the college Drone Team, adept at leading technical projects and integrating mechanical design with electronics for high-performance applications.

EXPERIENCE

CAD Intern <i>Coratia Techonologies Pvt Ltd</i>	June 2025 – Oct 2025 <i>NIT Rourkela , Odisha</i>
Developed an automated winch system for an underwater Remotely Operated Vehicle (ROV) with a 350m tether length, inclusive of comprehensive motion studies and simulations.	
Technical Head <i>Team Rotor FPV</i>	July 2024 – Sept 2025 <i>VIT, Vellore</i>
- Developed an autonomous drone with Level 2 autonomy, utilizing Ardupilot and the MATLAB UAV Toolbox. - Developed a 5inch FPV drone capable of carrying payloads upto 2kg and achieving speed of 200km/hr - Provided instruction in piloting and drone assembly to junior sophomores and participants in College Techfest - Engineered and published a patent for a 7-inch frame featuring a robust landing structure and reduced motor-induced vibration	
Member IPR-Cell <i>Institution's Innovation Council</i>	Sept 2024 – Dec 2025 <i>VIT, Vellore</i>
- Assisted students in the management and publication of their invention disclosure forms through VIT's IPR cell. - Prepared design reports and organized workshops and speaker sessions to spread awareness regarding intellectual property - Understood the norms of patent writing to facilitate a comprehensive literature review of inventions	
Intern Trainee <i>SAIL - RSP</i>	May 2025 – June 2025 <i>Rourkela , Odisha</i>
- Collaborated with experienced engineering executives to gain insights into shopfloor operations and maintenance department protocols - Gained in-depth knowledge regarding the application and operational principles of safety valves, as well as the procedures for conducting tank inspections across the entire plant. - A comprehensive understanding of the molten steel transfer process within industrial plants and the associated PLC systems was gained - Completed a project to analyze the significant hazards associated with the digitization of safety valves within the plant.	
Head of Events <i>Design Club (Pixelate)</i>	Jan 2024 – Jan 2025 <i>VIT, Vellore</i>
- Successfully managed over 700 students, organized workshops, and hackathons, and oversaw backend operations - An outreach event was organized for students in Classes 5 to 10, focusing on design and color theory.	

PROJECTS & ACHIEVEMENTS

Patent Application: 7inch UAV Frame with V legs (Application No: IN202641075421)	Published
Engineered a carbon-fiber composite quadcopter frame with integrated propeller guards and vibration-damping brace members to improve structural durability for reconnaissance and scouting applications.	
Patent Application: Rover Suspension (Application No: IN202441052602)	Published
- Engineered and filed a patent application for a novel rover suspension system redesign, which increased the rover's speed by 30% while significantly reducing pitching and rolling during operation. - Conducted extensive research and development, including advanced CAD modeling, dynamic simulations, and prototype testing to validate the innovative design concept and its performance benefits.	
International Space Drone Challenge(ISDC 2025), BITS Goa	1st Place
Secured 1st place by autonomously navigating cones while guiding a rover and collecting atmospheric data using a custom X-frame quadcopter.	
Guerin Tech Challenge, IANC VIT	1st Place
Awarded 1st place for prototyping a hardware solution for CubeSat deployment structures, demonstrating innovative design and functionality.	
Patent Writing Competition, VIT Vellore	1st Place
Secured 1st position in the IP Writing Competition organized by Institution's Innovation Council, VIT Vellore, showcasing expertise in intellectual property documentation.	
Hackwars, MeiTy and Startup Tamil Nadu	Best Hardware Implementation Prize
Received the Prize for an drone project for disaster management with telemetry upto 5km.	
Multirotor-Techkriti Drone Competition, IIT Kanpur	3rd Place
Designed and fabricated the drone's grabbing mechanism for an autonomous UAV mission to pick and drop a payload.	
AeroRush, NIT Trichy	2nd Place
Secured 2nd place in the overall competition utilizing a FPV drone.	

TECHNICAL SKILLS & CERTIFICATIONS

CAD/CAE Software: SolidWorks (CSWP-Mechanical Design, CSWPA-Drawing Tools, CSWPA-Sheet Metal, CSWA-Additive Manufacturing, CSWA-Sustainable Designing, 3Dswymer), Fusion 360
Simulation & Analysis: MATLAB & Simulink (Simulink Onramp, Simscape Onramp), Static Simulations
Programming & Control: Python, Java, ArduPilot, ROS
Manufacturing: Additive Manufacturing, Prototyping, Design for Manufacture and Assembly (DFMA)
Other Relevant Skills: Drone Piloting, Patent Writing, Technical Documentation, Project Planning, Management

EDUCATION

Vellore Institute of Technology <i>Bachelor of Technology in Mechanical Engineering CGPA - 8.86</i>	Vellore, Tamil Nadu <i>Aug 2022– July 2026</i>
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